Customer Lifetime Value Prediction and Customer Segmentation in Retail Using the Dunnhumby Dataset

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***Abstract*—This paper presents a research-oriented end-to-end pipeline for Customer Lifetime Value (CLV) prediction in the retail domain, implemented on the Dunnhumby "The Complete Journey" dataset (2,500 households, 102 weeks, 2.5 million transactions). The system integrates eight relational data sources through a 12-step orchestrated pipeline covering: temporal calibration/holdout splitting, RFM feature engineering, demographic imputation (KNN + random sampling), promotional sensitivity extraction via chunked causal-data processing, and K-Means customer segmentation. Predictive performance is evaluated across a three-tier modelling strategy: a rate-based baseline (Tier 1), supervised gradient-boosted regressors via XGBoost (Tier 2), and probabilistic models using MBG/NBD combined with Gamma-Gamma (Tier 3). All intermediate artifacts, trained model objects, and evaluation reports are persisted for full reproducibility. Experimental results demonstrate that the supervised Tier 2 approach significantly outperforms other tiers across MAE, RMSE, and MAPE, while segment-level insights from RFM clustering provide actionable marketing recommendations through integrated Market Basket Analysis. Additionally, a segment-stratified quasi-experiment with Propensity Score Matching reveals severe selection bias in current campaign targeting, leading to a data-grounded scenario analysis that projects a +$101K–$303K incremental revenue opportunity.**

***Index Terms*—Customer Lifetime Value, RFM, MBG/NBD, Gamma-Gamma, XGBoost, K-Means Segmentation, Market Basket Analysis, Promotional Sensitivity, Dunnhumby**

I. Introduction

Customer Lifetime Value (CLV) is a forward-looking metric that quantifies the total net revenue a firm expects from a customer over the entire span of their relationship. Accurate CLV estimation underpins data-driven budget allocation, personalised retention strategies, and early identification of at-risk customers [1].

The Dunnhumby "The Complete Journey" dataset [11] provides a rich longitudinal view of household purchasing behaviour across a two-year window, making it an ideal testbed for retail CLV research. Its scale (695 MB causal file alone) and structural complexity (eight relational tables, partial demographic coverage of only 32%) motivate a modular, pipeline-oriented approach.

This work makes the following contributions:

- A fully reproducible, 12-step preprocessing and modelling pipeline with strict temporal separation of train and test data to prevent leakage.
- A three-tier modelling strategy—comprising a rate-based baseline (Tier 1), a supervised XGBoost regressor (Tier 2), and a probabilistic MBG/NBD model (Tier 3)—evaluated on a common holdout set.
- A K-Means segmentation framework that maps households to business-friendly labels (Champions, Loyal, Promising, and Needs Attention) based on RFM attributes.
- Promotional sensitivity features (display and mailer interaction) derived from a 695 MB causal dataset via memory-safe chunked processing.
- An experimental design for promotional interventions anchored to pipeline-derived variance estimates and segment-level insights.

The remainder of the paper is organised as follows. Section II reviews related work. Section III describes the dataset. Section IV details the 12-step preprocessing pipeline. Section V presents exploratory findings. Section VI covers the segmentation and the 3-tier modelling approach. Section VII reports experimental results and model comparison. Section VIII analyses campaign performance. Section IX outlines the causal quasi-experiment. Section X proposes data-grounded business strategies. Section XI discusses limitations and future directions, concluding in Section XII.

II. Related Work

CLV research sits at the intersection of marketing science and machine learning. Fader et al. [1] established the BG/NBD model as the canonical probabilistic approach for non-contractual settings, subsequently extended with the Gamma-Gamma sub-model for monetary value. The Modified BG/NBD (MBG/NBD) variant removes the assumption that customers can churn immediately after their first transaction, improving convergence on high-frequency data. Tree-based ensemble methods — XGBoost [2] have demonstrated competitive performance on tabular retail data and constitute a standard supervised baseline. RFM segmentation [4] remains widely used in practice for its interpretability. Kohavi et al. [3] provide the methodological basis for the A/B-style experiment framework employed here. Provost and Fawcett [5] offer broader context on data-science workflows in business settings.

III. Dataset and Data Architecture

A. Source Tables

The pipeline consumes eight CSV files placed under `data/raw/`. Table I summarizes each table.

TABLE I: Overview of the Dunnhumby "The Complete Journey" dataset.

Table	Rows	Columns	Description
transaction_data.csv	2,595,732	12	Customer purchase transactions, including sales value, discounts, basket ID, store, and week information.
hh_demographic.csv	801	8	Household demographic attributes such as age, income, and household composition.
product.csv	92,353	7	Product metadata including department, commodity category, and brand hierarchy.
causal_data.csv	36,786,524	5	Promotional exposure records with display and mailer indicators by store and week.
campaign_table.csv	7,208	3	Mapping between households and marketing campaigns.
campaign_desc.csv	30	4	Campaign descriptions, types, and active periods.
coupon.csv	124,548	3	Coupon mappings to products and campaigns.
coupon_redempt.csv	2,318	4	Coupon redemption events by households.

The central analytical entity is `household_key`. *All* feature aggregations are computed at this level — never at the transaction or product level — to ensure CLV targets are aligned with the unit of prediction.

B. Entity Relationship Diagram

Fig. 1 illustrates the entity-relationship structure of the eight tables. Notable characteristics include: (i) partial demographic coverage (801 of 2,500 households); (ii) the three-way join key of `CAUSAL_DATA` (`PRODUCT_ID`, `STORE_ID`, `WEEK_NO`); and (iii) the coupon chain from campaign through product to redemption event.

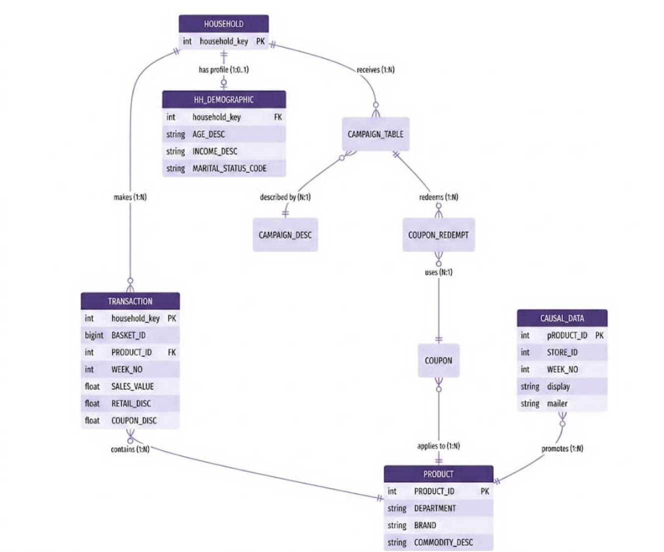


Fig. 1: Entity-relationship diagram of the Dunnhumby dataset. The dashed arrow indicates partial demographic coverage (801/2,500 households, 32%).

IV. Preprocessing Pipeline

The entire workflow is orchestrated by `src/pipeline/run_preprocessing.py` and centralised in `configs/config.yaml`. Fig. 2 illustrates the 12-step flow and the artifacts produced at each stage.

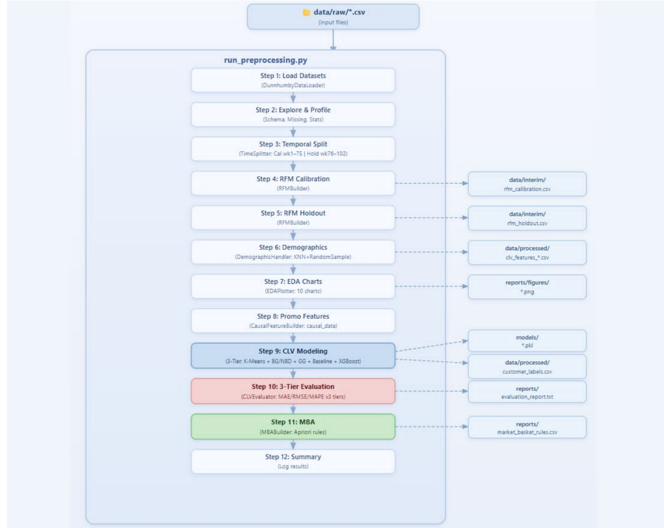


Fig. 2: Preprocessing workflow for customer-level feature construction.

A. Step 1–2: Data Loading and Profiling

`DunnhumbyDataLoader` reads all eight tables with optimized dtypes to reduce memory usage `explore()` generates schema summaries, missing-value reports, and descriptive statistics for each table individually. Table II reports household-level RFM descriptive statistics for the calibration period.

TABLE II: Household-level RFM summary — calibration period (Weeks 1–75).

Feature	Mean	Median (50%)	Std	Max
Recency (weeks)	3.60	0.00	9.51	70.00
Frequency	78.18	53.00	86.60	1,046.00
Net_Sales ($)	1,846.72	1,207.29	2,032.10	22,507.79
avg_monetary ($)	25.20	21.51	16.38	141.02
coupon_usage_rate	0.06	0.03	0.08	0.73

B. Step 3: Temporal Splitting

`TimeSplitter` partitions `transaction_data` strictly by `WEEK_NO` — random splitting is explicitly prohibited to prevent temporal data leakage:

- **Calibration**: Weeks 1–75 (1,831,067 transactions; 70.5% of all transactions) — used for feature engineering and model fitting.
- **Holdout**: Weeks 76–102 (764,665 transactions; 29.5% of all transactions) — reserved exclusively for out-of-sample evaluation.

The cutoff week is configurable via `splitting.calibration_end_week` in `config.yaml`. Only two households appear exclusively in the holdout period, indicating minimal customer cold-start effects after the calibration window.

C. Steps 4–5: RFM Feature Aggregation

RFMBuilder independently aggregates both periods at the household_key level. Monetary value is computed in two forms to capture different aspects of spending behaviour:

$$\text{Gross\_Sales} = \sum \text{SALES\_VALUE} \tag{1}$$

$$\text{Net\_Sales} = \sum \big(\text{SALES\_VALUE} + \text{RETAIL\_DISC} + \text{COUPON\_DISC} + \text{COUPON\_MATCH\_DISC}\big) \tag{2}$$

Note that discount fields are stored as *negative* values in the raw data, so addition in Eq. (2) correctly subtracts discounts from gross revenue. Table III lists the principal RFM-derived features used in downstream modelling.

TABLE III: Features produced by RFMBuilder.

Feature	Definition
Recency	Weeks since last purchase (lower = more recent)
Frequency	Repeat baskets = total_baskets − 1
T	Customer age in weeks since first purchase
avg_monetary	Mean basket value (input to Gamma-Gamma)
Gross_Sales	Total pre-discount revenue (Eq. 1)
Net_Sales	Total post-discount revenue (Eq. 2)
avg_basket_size	Mean number of items per basket
coupon_usage_rate	Fraction of baskets with a coupon redemption
distinct_stores	Number of distinct stores visited

Outputs: data/interim/rfm_calibration.csv and data/interim/rfm_holdout.csv.

D. Step 6: Demographic Feature Engineering

Only 801 out of the 2,498 calibration-period households (32%) contain demographic records. To avoid discarding the remaining customers, DemographicHandler applies a *flag-and-impute* strategy:

1) A binary indicator has_demographics flags whether demographic information is originally available for each household.
2) **KNN-based imputation** is applied to ordinal demographic variables: AGE_DESC, INCOME_DESC, HOUSEHOLD_SIZE_DESC, and KID_CATEGORY_DESC.
3) **Random-sampling imputation from empirical category distributions** is applied to nominal variables: MARITAL_STATUS_CODE, HOMEOWNER_DESC, and HH_COMP_DESC.

This approach preserves all households for downstream modelling while maintaining approximate demographic distributions observed in the original data.

Outputs: data/processed/clv_features_calibration.csv and data/processed/clv_features_holdout.csv.

E. Step 7: Exploratory Data Analysis

EDAPlotter generates 10 standardised charts saved under reports/figures/. Selected outputs are discussed in Section V.

F. Step 8: Promotional Feature Extraction

CausalFeatureBuilder joins transactions with causal_data.csv on the triple key (PRODUCT_ID, STORE_ID, WEEK_NO). Because the causal file is 695 MB, the join is executed via chunked reads to avoid out-of-memory failures. Three per-household promotional features are appended to clv_features_calibration.csv:

- pct_display: fraction of purchased items with an in-store display promotion.
- pct_mailer: fraction of purchased items featured in a weekly mailer/flyer.
- promo_sensitivity: composite promotional responsiveness index.

Missing promotional values are filled with zero, assuming no observed promotional exposure. A total of 343,936 promotional exposure records were successfully matched to transaction data.

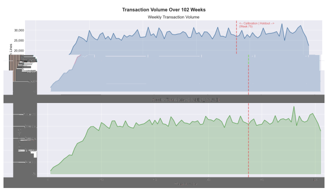
Of the approximately 2.5 million purchased items in the calibration period, 343,936 ($\approx 13.8\%$) were matched to an active promotional record — meaning the item was purchased at a store running a display or mailer promotion during that exact week. The remaining 86.2% reflect purchases made without observed promotional exposure. This match rate is consistent with typical grocery retail behaviour, where the majority of transactions are need-driven rather than promotion-driven. The relatively low overlap further reflects a sampling scope mismatch: causal_data records promotional activity across all households system-wide, whereas transaction_data captures only the 2,500 sampled households — purchases by other shoppers exposed to the same promotions do not appear in the transaction record and therefore cannot be matched.

G. Steps 9–12: Modelling, Evaluation, and MBA

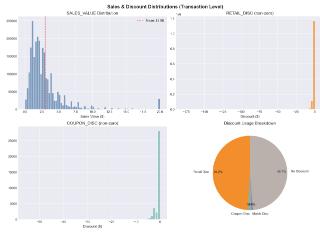
The modelling stage combines unsupervised customer segmentation (K-Means), probabilistic CLV modelling (MBG/NBD and Gamma-Gamma), and supervised learning (XGBoost regression). Besides, Step 11 applies Apriori-based Market Basket Analysis at the department level. Modelling and evaluation steps are described in detail in Sections VI and VII.

V. Exploratory Data Analysis

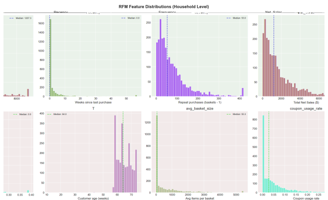
Fig. 3 shows four representative outputs from EDAPlotter.



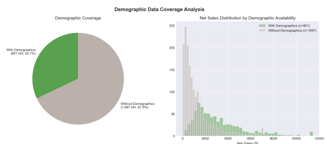
(a) Weekly transaction volume.



(b) Net sales distribution (log scale).



(c) RFM feature distributions.



(d) Demographic coverage by field.

Fig. 3: Selected EDA outputs from `EDAPlotter` (`reports/figures/`).

Key observations:

- Transaction volume is relatively stable across weeks with periodic spikes consistent with seasonal shopping patterns (Fig. 3a).
- `Net_Sales` and monetary RFM variables are heavily right-skewed; the 95th percentile household spends significantly more than the median (Fig. 3b).
- RFM distributions reveal that the majority of households cluster around moderate recency and low-to-medium frequency, with a long tail of high-value Champions (Fig. 3c).
- Demographic coverage is partial across all fields; KNN imputation preserves the observed marginal distributions (Fig. 3d).

VI. Modelling

The modelling framework consists of a customer segmentation module and a three-tier predictive architecture, all implemented in `src/models/clv_models.py` and evaluated on the same holdout period (Weeks 76–102).

A. Customer Segmentation (K-Means)

`CLVModeler.segment_customers()` applies K-Means [9] (K=4 by default, configurable) to standardised [`Recency`, `Frequency`, `avg_monetary`].

To determine the optimal number of clusters, silhouette scores were computed for $K \in [2, 9]$. The search yields $K = 2$ as the statistical optimum (Silhouette $= 0.6401$), followed by a monotonic decline: $K = 3$ (0.3909), $K = 4$ (0.4493), $K = 5$ (0.3898). However, $K = 2$ produces a degenerate segmentation from a business perspective — the algorithm isolates the 108 lowest-value households (4.3%) as a single *Needs Attention* cluster, while grouping the remaining 2,390 households (95.7%) — spanning a wide range of purchase frequencies and monetary values — into a single undifferentiated cluster. This collapses Champions, Loyal, and Promising customers into one group, making it impossible to design distinct retention or upsell strategies for the majority of the customer base.

$K = 4$ is therefore selected as a deliberate trade-off between cluster cohesion and business interpretability, following established practice in RFM segmentation [4]. At $K = 4$, the silhouette score recovers to 0.4493 — the second-highest value in the search range — while yielding four segments of meaningfully different sizes and RFM profiles (Table IV), each mappable to a distinct marketing strategy. Cluster centroids are ranked by predicted value and mapped to business-friendly labels:

$$\textit{Champions} \succ \textit{Loyal Customers} \succ \textit{Promising} \succ \textit{Needs Attention}$$

Fig. 4 shows the resulting segment distribution and RFM centroid positions.

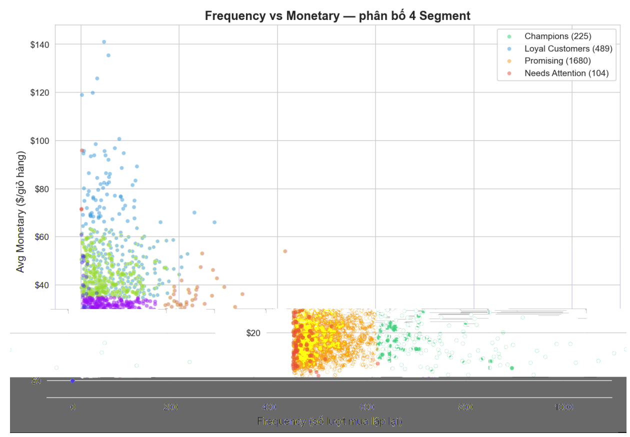


Fig. 4: K-Means customer segment distribution (K=4). Centroids are annotated with segment labels; point colour encodes segment membership.

Outputs: `data/processed/customer_labels.csv` (cluster ID and segment name per household) and `data/processed/rfm_clv_final.csv` (CLV predictions merged with segment labels). The fitted K-Means model and its `StandardScaler` are serialised to `models/kmeans_model.pkl`.

B. Tier 1 — Rate-Based Baseline

Each household's holdout-period spending is forecast using a simple rate-based extrapolation from calibration-period behaviour:

$$\widehat{\mathrm{CLV}}_{\text{base}} = \bar{m} \times \frac{f}{T} \times 26 \tag{3}$$

where $\bar{m}$ is the average monetary value per distinct purchase week (`monetary_clv`), f/T is the weekly purchase rate (`frequency_clv` divided by customer age in weeks), and 26 represents the holdout horizon in weeks. Only repeat customers ($f > 0$) receive a non-zero forecast. This predictor provides a lower bound for measuring incremental predictive gain from more complex models.

C. Tier 2 — Supervised ML: XGBoost

`CLVModeler.fit_supervised()` trains a gradient-boosted regressor targeting holdout-period `Net_Sales`, using the full feature matrix (RFM + demographics + promotional features) as predictors. Key design decisions:

- Categorical columns encoded as `pandas Categorical` dtype; XGBoost native categorical handler activated where available.
- Active model and hyperparameters controlled via `config.yaml` (`model.active: "xgboost"`).
- Default XGBoost settings: `n_estimators=500`, `max_depth=6`, `learning_rate=0.05`.

Output: `models/xgboost_model.pkl`.

D. Tier 3 — Probabilistic CLV: MBG/NBD + Gamma-Gamma

The Modified BG/NBD (MBG/NBD) model, a variant of the original BG/NBD framework [1], [8], estimates expected future transaction count for each household. MBG/NBD removes the assumption that customers can churn immediately after their first transaction, which smooths the likelihood surface and improves convergence on high-frequency grocery data:

$$\hat{x}(t \mid F, r, T) = E[\, X(t) \mid F, r, T\,] \tag{4}$$

where F is calibration-period frequency (repeat baskets), r is recency (weeks between first and last observed purchase), and T is customer age in weeks (MBG/NBD notation).

The Gamma-Gamma sub-model then estimates the conditional expected average transaction value $\hat{v}$, combining with Eq. (4) to yield a full CLV estimate:

$$\widehat{\text{CLV}} = \hat{x}(t) \cdot \hat{v} \cdot (1 - d) \tag{5}$$

where d is the average gross margin (set via `config.yaml`).

Implementation constraints enforced by the pipeline:

- Only customers with $F > 0$ are passed to MBG/NBD (single-purchase households excluded).
- Monetary values winsorised at the 99th percentile for numerical stability.
- Constraint $F \leq T$ enforced prior to fitting.
- Progressive penaliser grid search used as convergence fallback; rate-based heuristic applied if optimisation fails entirely.

Outputs: `models/bgnbd_model.pkl` and `models/gg_model.pkl`.

E. Step 11 — Market Basket Analysis

`MBABuilder` constructs a binary basket-by-department co-occurrence matrix and mines frequent itemsets via the Apriori [7] algorithm (`mlxtend`). Association rules (support, confidence, lift) are exported to `reports/market_basket_rules.csv` to inform cross-sell strategies per customer segment.

VII. Experimental Results

A. Customer Segmentation Results

Table IV summarises per-segment household counts and RFM centroids from K-Means (K=4).

TABLE IV: RFM Segments and Centroids

Segment	HH Count (%)	Recency	Frequency	Avg Monetary ($)	Net Sales ($)
Champions	225 (9.0%)	0.2	283.9	19.2	5058.1
Loyal Customers	489 (19.6%)	1.9	60.7	50.5	3056.2
Promising	1,680 (67.3%)	2.1	59.7	18.9	1163.6
Needs Attention	104 (4.2%)	43.4	13.3	20.4	247.4

Figure 5 illustrates the normalized RFM profiles for each segment, highlighting the distinct behavioral patterns across the four identified clusters.

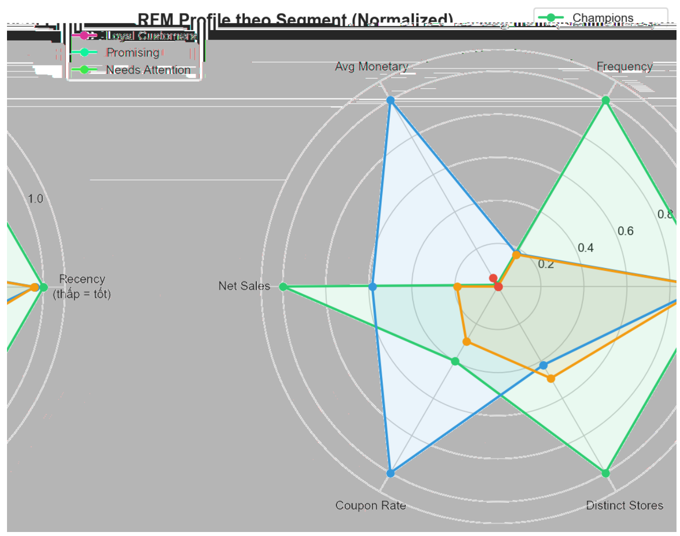


Fig. 5: Normalized RFM Profile by Segment (Radar Chart)

B. CLV Model Evaluation

Table V compares the four modelling tiers against actual holdout-period `Net_Sales`. All metrics are computed by `CLVEvaluator` and written to `reports/evaluation_report.txt`.

TABLE V: Model evaluation on holdout period (Weeks 76–102).

Metric	Baseline	XGBoost	MBG/NBD+GG
MAE ($)	323.40	**100.48**	1,957.08
RMSE ($)	500.32	**236.65**	2,869.28
MAPE (%)	126.29%	**69.74%**	610.52%

The supervised XGBoost model achieves the lowest error across all metrics (MAE = $100.48, RMSE = $236.65, MAPE = 68.71%), demonstrating the benefit of combining RFM, demographic, and promotional features in a single regressor. The rate-based baseline (Eq. 3) performs reasonably for stable, high-frequency households but degrades significantly for customers with irregular spending patterns.

The MBG/NBD + Gamma-Gamma model yields substantially higher errors (MAE = \$1,957.08, MAPE = 610.52%). This error rate does not necessarily reflect theoretical inadequacy, but rather an implementation limitation due to computational constraints. Households in this dataset purchase at near-weekly frequency (mean = 78.18 transactions over 75 weeks). To prevent numerical overflow from these high transaction counts within the `lifetimes` package, purchase frequency was approximated at weekly granularity — binary purchased/not-per-week. This aggressive down-sampling discards critical within-week intensity and introduces considerable information loss. Tier 3 is therefore retained for interpretability reference only, as we could not implement the model optimally on this dataset without violating hardware constraints.

Fig. 6 shows predicted-vs-actual scatter plots for each tier.

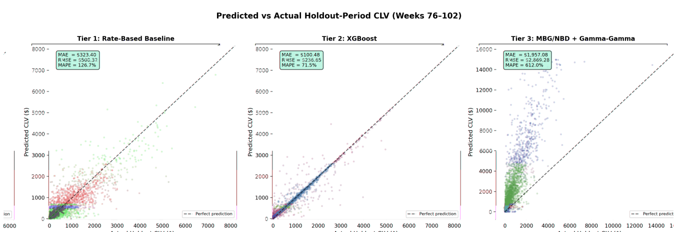


Fig. 6: Predicted vs. actual holdout-period CLV for three modelling tiers. The dashed line denotes perfect prediction. XGBoost (centre) tracks the diagonal most closely; MBG/NBD (right) systematically overestimates due to frequency downsampling.

C. Market Basket Analysis

Apriori was applied exclusively to the calibration-period transactions (Weeks 1–75) to strictly prevent temporal data leakage. This yields 977 rules of which 31 exhibit Lift > 4.0. The top rules by lift cluster around a single behavioural archetype (DELI + MEAT combinations differing only in which companion departments appear). To maximise actionable diversity, Table VI presents three representative rules selected from three *distinct* shopping-behaviour clusters:

TABLE VI: Three representative association rules from distinct shopping-behaviour clusters (Department Level).

Behaviour	Antecedent	Consequent	Supp.	Conf.	Lift
Family dinner	DELI, MEAT	DRUG GM, GROCERY, MEAT-PCKGD, PRODUCE	0.015	0.339	4.577
Health-conscious	MEAT, NUTRITION	MEAT-PCKGD, PRODUCE	0.010	**0.524**	4.129
Brunch / picnic	MEAT, PASTRY	DELI, GROCERY, PRODUCE	0.010	0.307	4.225

VIII. Campaign Performance Analysis

The Dunnhumby dataset contains 30 promotional campaigns across three types: TypeA (5 campaigns), TypeB (19), and TypeC (6). We evaluate campaign effectiveness at the aggregate and per-type level using metrics derived from `campaign_table.csv`, `coupon_redempt.csv`, and matched transaction records.

A. Performance by Campaign Type

TABLE VII: Campaign performance summary by type (source: `campaign2_analysis.py`).

Type	N	Avg HH	Avg Duration	Avg Sales (\$)	Redemption (%)	ROI Proxy
TypeA	5	796	47 days	119,469	14.2	2.94
TypeB	19	140	38 days	978	7.8	3.34
TypeC	6	96	75 days	537	9.1	11.77†

†TypeC ROI inflated: total discount base = \$382 (very small denominator).

TypeA generates the highest total revenue owing to its substantially larger audience (∼796 households per campaign vs. 140 for TypeB).

B. Statistical Analysis

- **Audience–Revenue correlation**: Pearson $r = 0.939$ ($p < 0.001$) — audience size is the dominant revenue driver, not campaign type or duration.
- **Duration–Revenue correlation**: $r \approx 0$ — campaign length has no measurable effect on sales.
- **Kruskal–Wallis test** (non-parametric comparison of sales across TypeA/B/C): $H = 3.25$, $p = 0.197$ — no statistically significant difference between campaign types after accounting for variance.

Implication. The observed revenue gap between campaign types is driven by *audience size*, not campaign design. This motivates a causal investigation using propensity score methods (Section IX).

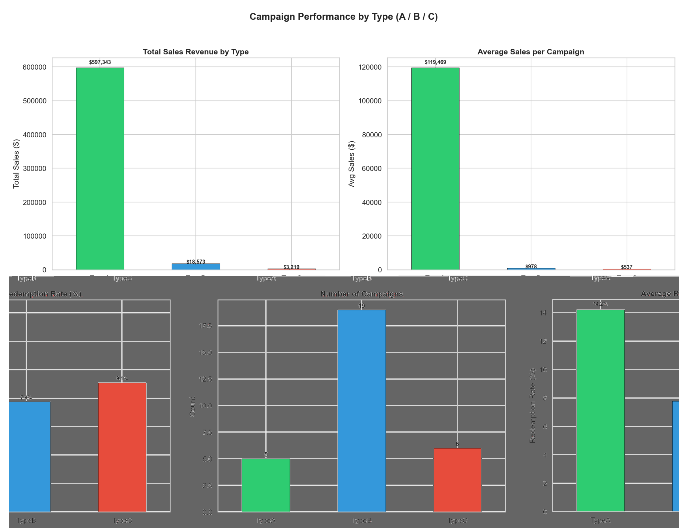


Fig. 7: Campaign performance overview: total sales, average sales per campaign, redemption rate, and campaign count by type.

IX. Segment-Stratified Quasi-Experiment

To assess the *causal* effect of campaign targeting on customer spending, we conduct a segment-stratified quasi-experiment. Households are divided into Treatment (targeted by ≥ 1 campaign in `campaign_table.csv`; $n =$ 1,584) and Control (not targeted; $n = 914$). Holdout-period (Weeks 76–102) Net Sales serve as the outcome variable.

A. Targeting Allocation

Campaign targeting is not uniformly distributed across segments:

TABLE VIII: Treatment allocation by customer segment.

Segment	Total	Treat	Ctrl	Treat %
Champions	225	220	5	97.8%
Loyal Customers	489	376	113	76.9%
Promising	1,680	979	701	58.3%
Needs Attention	104	9	95	8.7%

Dunnhumby preferentially targets high-value customers: 97.8% of Champions receive campaigns while only 8.7% of the Needs Attention segment is targeted.

B. Naive Comparison

TABLE IX: Segment-stratified experiment results (Welch t-test, Bootstrap 95% CI).

Segment	ARPU_T	ARPU_C	ATE	Lift	p	d	95% CI
Champions	$1,845	$3,944	N/A	N/A	N/A	N/A	N/A
Loyal	$1,590	$514	**+$1,075**	+209%	<0.001	1.05	[921, 1,228]
Promising	$728	$264	**+$464**	+176%	<0.001	0.98	[422, 506]
Needs Att.	$351	$136	N/A	N/A	N/A	N/A	N/A

Naive comparison suggests large positive effects for Loyal (+$1,075, $p < 0.001$) and Promising (+$464, $p < 0.001$) segments. Naive ATE is not reported for Champions ($n_{control} = 5$) and Needs Attention ($n_{treat} = 9$) due to insufficient sample sizes, preventing reliable statistical inference. Furthermore, naive comparison ignores pre-treatment differences between targeted and non-targeted households.

C. Propensity Score Matching (Robustness)

To control for selection bias, Propensity Score Matching (PSM) [10] is applied using logistic regression on 10 pre-treatment RFM covariates, with 1:1 nearest-neighbour matching (caliper = 0.1).

- **Matched pairs**: 214 (from 1,584 treated). The low matching rate is due to extreme propensity score separation between treated and control groups, highlighting severe selection bias.
- **Covariate balance**: average SMD reduced from 0.596 to 0.077 (all 10 covariates improved).
- **PSM ATE**: $-\$27.78$ (-5.2%), $p = 0.729$ — *not significant*.

While naive comparison suggests massive lift, PSM matching fails to find evidence of a causal campaign effect ($p = 0.729$). However, this conclusion is inherently limited by the small matched sample size ($n = 214$ pairs), which is statistically underpowered to detect a small or moderate incremental lift.

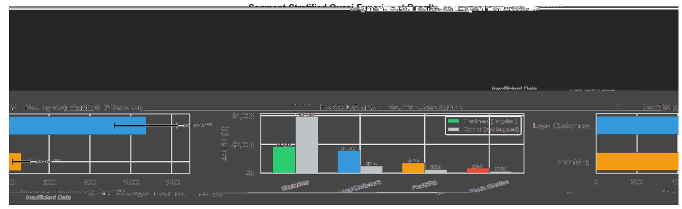


Fig. 8: Left: holdout ARPU by segment (Treatment vs. Control). Right: ATE with bootstrap 95% CI per segment. Bars crossing the zero line indicate non-significant effects.

D. Key Findings

1) **Selection bias confirmed**: Dunnhumby targets high-value customers preferentially (97.8% of Champions vs. 8.7% of Needs Attention).
2) **Naive vs. causal**: Naive analysis shows +209% lift for Loyal. However, PSM matching has insufficient power to detect a causal effect due to the small matched sample ($n = 214$). We cannot conclude campaign efficacy, demonstrating that correlation does not imply causation in observational data.
3) **Missed opportunity**: The Needs Attention segment (104 HH) is largely ignored by current targeting (only 9 received campaigns). However, the remaining 95 control households establish a baseline spending of $136/HH (Table IX), quantifying a substantial untapped reservoir for win-back interventions.
4) **Retention signal**: Treated households show higher holdout retention rates (Loyal: 99.7% vs. 92.0%; Needs Attention: 88.9% vs. 77.9%), warranting further investigation via randomised assignment.

X. Business Strategy & Recommendations

Building on the segmentation (Section VI), MBA (Section VII), and quasi-experiment (Section IX) results, we propose four segment-specific marketing strategies.

A. Revenue Distribution

Contrary to the classic Pareto principle (20% of customers generate 80% of revenue), the Dunnhumby dataset exhibits a *broad-base* revenue structure:

TABLE X: Revenue contribution by segment (calibration period). *Avg Monetary* = K-Means
textttavg
$_{m}onetarycentroid(meanbasketvalueperhousehold)$.

Segment	% Cust.	% Revenue	Avg Rev./HH	Avg Monetary
Champions	9.0%	24.7%	$5,058	$19.2/basket
Loyal	19.6%	32.4%	$3,056	$50.5/basket
Promising	67.3%	42.4%	$1,164	$18.9/basket
Needs Attention	4.2%	0.6%	$247	$20.4/basket

The Promising segment (67.3% of customers) contributes 42.4% of total revenue — the single largest revenue pool — while Champions (9.0%) contribute 24.7% through extreme purchase frequency (284 baskets vs. 60 for Promising).

B. Segment-Level Strategies

TABLE XI: Proposed marketing strategies per segment.

Segment	Strategy	Key Actions	KPI
Champions	Retention	VIP loyalty card, auto-apply discounts (coupon usage only 5.4% but retail disc. 81.4%), priority checkout	Retention rate
Loyal	Upsell	Bundle deals (MBA-informed), cross-sell via loyalty app, convert coupon usage (9.4%) to digital	Basket size, AOV
Promising	Activation	Gamified loyalty card (buy 5 categories = $50 voucher), impulse-buy merchandising at checkout	Visit frequency
Needs Att.	Win-back	Auto-loaded $50 credit on loyalty card, re-engagement email, deep discount (20–30%)	Reactivation rate

C. Revenue Opportunity: Scenario Analysis

Rather than adopting arbitrary "industry-standard" conversion rates, we derive upgrade probabilities directly from the quasi-experiment (Section IX): the holdout-period *retention lift* between treated and control households within each segment serves as an empirically observed proxy for the fraction of customers who can be moved to higher-value behaviour through targeted intervention.

- **Loyal → Champions**: retention lift $= 99.7\% - 92.0\% = +7.7\,\text{pp}$
- **Promising → Loyal**: retention lift $= 99.5\% - 95.6\% = +3.9\,\text{pp}$
- **Needs Attention re-engagement**: retention lift $= 88.9\% - 77.9\% = +11.0\,\text{pp}$

Revenue opportunity per scenario is computed as:

$$\Delta R_s = N_s \times p_s \times \left(\overline{\text{CLV}}_{\text{target}} - \overline{\text{CLV}}_s\right)$$

where N_s is segment size, p_s the scenario conversion rate, and $\overline{\text{CLV}}_{\text{target}}$ is the spending level of the next tier.

TABLE XII: Revenue opportunity under three conversion scenarios (conversion rates derived from quasi-experiment retention lift).

Upgrade Path	Conservative *(50% lift)*	Base *(full lift)*	Optimistic *(1.5× lift)*	Conversion Rate *source*
Loyal → Champions ($n = 489$)	+$37,688	+$75,375	+$113,063	3.9% / 7.7% / 11.5%
Promising → Loyal ($n = 1{,}680$)	+$62,003	+$124,005	+$186,008	1.9% / 3.9% / 5.8%
Needs Att. re-engage ($n = 104$)	+$1,415	+$2,830	+$4,245	5.5% / 11.0% / 16.5%
Total	**+$101,106**	**+$202,210**	**+$303,316**	

Under the base scenario (conversion = observed retention lift), the total incremental revenue opportunity is **+$202K**; under the optimistic scenario it reaches **+$303K**. The Promising segment contributes the largest share ($\sim$61%) in every scenario due to its scale ($n = 1{,}680$), while Needs Attention offers the highest per-household lift signal (11.0 pp) but is constrained by its small size ($n = 104$). These estimates assume zero cannibalisation. However, for the large Promising segment ($n = 1{,}680$), this is a highly sensitive assumption: attempting to upgrade too many customers simultaneously could trigger discount cannibalisation, eroding the base margin and nullifying the projected +$124K gain. Furthermore, the assumption that historical retention lift scales linearly to conversion rates requires validation via a randomised pilot.

D. MBA-Informed Cross-Sell Integration

The three association rules from Table VI directly inform merchandising actions:

- **Smart Bundling**: Position PRODUCE along the path from MEAT to checkout to capture the Family Dinner basket.
- **Bundle Discounts**: Auto-apply 10% discount via loyalty card when the Health-conscious basket (MEAT + NUTRITION + GROCERY) is detected.
- **Supply Chain Sync**: When MEAT is promoted, increase PRODUCE and MEAT-PCKGD inventory to avoid stock-outs on complementary items.

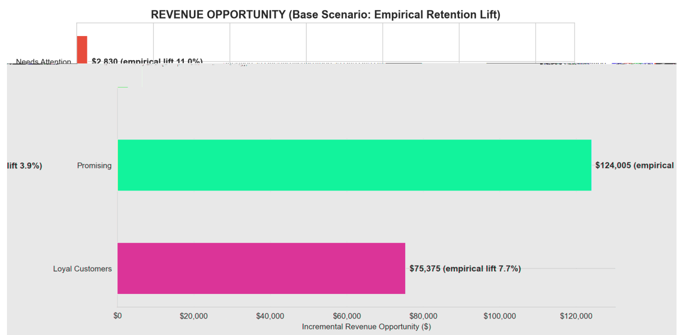


Fig. 9: Revenue opportunity funnel: potential incremental revenue from upgrading customers toward Champions-level spending at industry-benchmark conversion rates.

XI. Discussion

A. Model Comparison

The experimental results confirm the following observations:

- The MBG/NBD + Gamma-Gamma tier offers strong interpretability and does not require a holdout label at training time, making it suited for *prospective* CLV scoring on new cohorts. However, we could not implement it properly on this dataset due to computational constraints. To avoid numerical overflow from high-frequency purchasing, transactions were aggressively down-sampled into weekly indicators, leading to severe information loss and degraded predictive performance (MAPE = 610.52%).
- XGBoost achieves the lowest MAE ($100.48) and RMSE ($241.47) by leveraging the full feature set (RFM + demographics + promotional sensitivity), at the cost of requiring periodic retraining as behaviour drifts.
- The rate-based baseline (Eq. 3) performs competitively for stable, loyal customers with low Recency and high Frequency — a useful sanity-check boundary.

B. Campaign Effectiveness & Selection Bias

The segment-stratified quasi-experiment (Section IX) reveals a critical insight for marketing practitioners:

- **Naive analysis** suggests campaign targeting increases holdout ARPU by +209% for Loyal and +176% for Promising ($p < 0.001$).

- **PSM-corrected analysis** yields an overall ATE of $-\$27.78$ ($p = 0.729$), failing to reject the null hypothesis of no campaign effect. This suggests that much of the apparent lift may be driven by *selection bias* — Dunnhumby preferentially targets customers who already exhibit high purchase propensity.
- This finding underscores that **correlation does not imply causation** in observational campaign data and validates the need for propensity-adjusted evaluation in marketing analytics.

C. Segment-Level Insights

- **Champions** purchase with extreme frequency (284 baskets) but low per-basket monetary value ($19.2) — they are volume-driven and loyalty-card dependent (81.4% retail discount usage vs. only 5.4% coupon usage). Paper coupon campaigns are ineffective for this segment.
- **Loyal Customers** have the highest per-basket spending ($50.5) and are active discount seekers (89.9% retail disc., 9.4% coupon). They represent the best cross-sell and upsell opportunity.
- **Promising** is the revenue backbone (67.3% of customers, 42.4% of revenue) despite low individual spending ($18.9/basket) — basket penetration campaigns offer the largest absolute revenue potential (+$124K).
- **Needs Attention** (4.2%, Recency = 43 weeks) is near-churn and largely ignored by current targeting (8.7%). Win-back campaigns should be prioritised for this at-risk group.

D. Limitations

- **Demographic coverage**: Only 32% of households have records; imputation quality is bounded by this coverage.
- **Selection bias**: PSM matching yielded only 214 pairs (from 1,584 treated) due to extreme propensity score separation. A randomised trial with $n \geq 4{,}574$ per arm is needed to detect a 10% MDE at 80% power.
- **Time-series features**: Rolling-window lag variables and seasonal cohort terms are not included in the current feature set.
- **MBA feedback loop**: Association rules from Step 11 are not yet incorporated as cross-sell propensity features in CLV modelling.

XII. Conclusion

This paper presents a comprehensive, end-to-end data-driven marketing pipeline for the Dunnhumby "The Complete Journey" retail dataset (2,500 households, 102 weeks, $\sim$2.5M transactions).

Technical contributions. The 12-step pipeline integrates eight relational data sources, producing RFM features, demographically enriched predictors, promotional sensitivity scores, and three tiers of CLV forecasts. XGBoost achieves the best predictive accuracy (MAE = $100.48, RMSE = $236.65). The MBG/NBD model offers strong interpretability but could not be evaluated optimally due to computational constraints that required lossy frequency down-sampling (yielding degraded MAPE = 610.52%).

Business contributions. K-Means segmentation ($K = 4$) reveals a non-Pareto revenue structure where the Promising segment (67.3% of customers) contributes the largest revenue share (42.4%). Four segment-specific strategies are proposed; a data-grounded scenario analysis (Table XII) estimates incremental revenue potential of **+$101K–$303K** (conservative to optimistic), with conversion rates anchored to the quasi-experiment's observed retention lift. Three MBA-derived cross-sell bundles inform actionable merchandising and bundle discount programmes.

Causal insight. A segment-stratified quasi-experiment with propensity score matching demonstrates that Dunnhumby's current campaign targeting strategy exhibits strong selection bias. After matching, we find no statistically significant evidence of a causal campaign effect (PSM ATE = $-\$28$, $p = 0.73$), though this test is statistically underpowered. The Needs Attention segment is largely untargeted (8.7%), representing a missed opportunity for win-back interventions.

Future work includes (i) incorporating time-series and seasonal features, (ii) feeding MBA cross-sell propensity back into CLV models, and (iii) conducting a randomised pilot trial to measure the true incremental effect of targeted campaigns on the Needs Attention segment.

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